Formal Stability Bounds for RandNLA Algorithm 2

LeanFpAnalysis repository

Abstract

This note states, in standard mathematical form, the Lean formalization of Algorithm 2 from Drineas and Mahoney's CACM RandNLA survey, RandNLA: Randomized Numerical Linear Algebra. The first part records the row-sampling Frobenius guarantee for the squared-row-norm distribution. The second part specializes the same formal machinery to leverage-score sampling, equation (6), and proves the subspace-embedding estimate of equation (7), including the additional floating-point perturbation term.

1 Notation and Algorithm

Let $A \in \mathbb{R}^{m\times n}$. For its i-th row write A_{i*}, and set

$$r_i(A) = \|A_{i*}\|_2^2 = \sum_{j=1}^{n} A_{ij}^2, \qquad D(A) = \|A\|_F^2 = \sum_{i=1}^{m} r_i(A).$$

Algorithm 2 samples indices $I_1, \dots, I_s$ independently from a row distribution $p = (p_1, \dots, p_m)$. The sampled and rescaled sketch is

$$\widetilde{A}_{tj} = \frac{A_{I_t j}}{\sqrt{s\, p_{I_t}}}, \qquad 1 \le t \le s, \quad 1 \le j \le n.$$

The exact sampled Gram matrix is

$$G(\widetilde{A}) = \widetilde{A}^T \widetilde{A}, \qquad G(A) = A^T A.$$

The floating-point model has unit roundoff u. The repository uses γ_s for the standard dot-product error factor, with the Lean side-condition `gammaValid fp s`. We distinguish two computed Gram matrices:

$$\widehat{G}_{\text{scale}} = G(\widehat{\widetilde{A}}), \qquad \widehat{G}_{\text{dot}} = \mathrm{fl}_{\text{dot}}(\widehat{\widetilde{A}})^T\, \mathrm{fl}_{\text{dot}}(\widehat{\widetilde{A}}),$$

where $\widehat{\widetilde{A}}$ denotes the sketch obtained after rounded row scaling, and $\widehat{G}_{\text{dot}}$ computes each Gram entry using the repository's floating-point dot-product algorithm.

2 Equation (4): Squared-Row-Norm Sampling

Definition 1 (Squared-row-norm probabilities)**.** Assume $D(A) > 0$. The equation (4) distribution is

$$p_i = \frac{r_i(A)}{D(A)} = \frac{\sum_j A_{ij}^2}{\sum_{k,\ell} A_{k\ell}^2}.$$

Lemma 2 (Probability and support). *If $D(A) \neq 0$, then $\sum_i p_i = 1$. If $D(A) > 0$, then $p_i \geq 0$ for all i. Moreover, if row i contains a nonzero entry, then $p_i > 0$, and for $s > 0$ the scaling denominator $\sqrt{s\, p_i}$ is nonzero.*

Lean formalization.

`rowSqNormProb_sum_eq_one`

`rowSqNormProb_nonneg`

`rowSqNormProb_pos_of_entry_ne_zero`

`rowSampleScaleDen_ne_zero`

Lemma 3 (One rounded sampled row entry). *For a sampled row with $p_{I_t} > 0$, the rounded scaling operation satisfies*

$$\left|\widehat{\widetilde{A}}_{tj} - \widetilde{A}_{tj}\right| \leq u\, |\widetilde{A}_{tj}|.$$

Lean formalization.

`fl_rowSampleSketch_error_bound`.

Remark. Algorithm 2 does not aggregate repeated samples into a single row. Repeated draws produce repeated output rows. Consequently the relevant random object is the sampled Gram matrix $\widetilde{A}^T \widetilde{A}$, not a grouped hit-count update.

3 Exact Gram Approximation

Theorem 4 (Unbiased sampled Gram entries). *Assume $D(A) > 0$ and $s > 0$. Under the independent row sampler with probabilities $p_i = r_i(A)/D(A)$,*

$$\mathbb{E}\big[(\widetilde{A}^T \widetilde{A})_{jk}\big] = (A^T A)_{jk} \qquad \text{for all } j, k.$$

Lean formalization.

`rowSqNormTraceProbability_expectationReal_rowSampleGram_entry`

Theorem 5 (Second moment for the Frobenius error). *Assume $D(A) > 0$ and $s > 0$. Then*

$$\mathbb{E}\left[\left\|\widetilde{A}^T \widetilde{A} - A^T A\right\|_F^2\right] \leq \frac{D(A)^2}{s}.$$

Lean formalization.

`rowSqNormTraceProbability_expectationReal_sampleAverage_sub_mean_sq`

`rowSqNormTraceProbability_expectationReal_rowSampleGram_frob_error_sq_le`

Theorem 6 (Equation (5)). *Assume $D(A) > 0$, $s > 0$, and $\varepsilon > 0$. Then*

$$\mathbb{P}\left[\left\|\widetilde{A}^T \widetilde{A} - A^T A\right\|_F \leq \varepsilon\, D(A)\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Equivalently, since $D(A) = \|A\|_F^2$,

$$\mathbb{P}\left[\left\|\widetilde{A}^T \widetilde{A} - A^T A\right\|_F \leq \varepsilon\, \|A\|_F^2\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Lean formalization.

`FiniteProbability.eventProb_le_ge_one_sub_expectationReal_sq_div`

`rowSqNormTraceProbability_eventProb_rowSampleGram_frob_error_le_epsilon`

4 Floating-Point Perturbation of Equation (5)

The formalized perturbation bounds are deterministic on the probability-one support where every sampled row has positive sampling probability. Define $\mathbf{1}_{n\times n}$ to be the $n \times n$ all-ones matrix. The Lean budgets are Frobenius norms of constant $n \times n$ matrices:

$$\tau_{\text{scale}}(A) = \left\|(2u + u^2)D(A)\mathbf{1}_{n\times n}\right\|_F,$$

$$\tau_{\text{dot}}(A) = \left\|\gamma_s(1+u)^2 D(A)\mathbf{1}_{n\times n}\right\|_F, \qquad \tau_{\text{full}}(A) = \tau_{\text{scale}}(A) + \tau_{\text{dot}}(A).$$

Equivalently, since these scalar entries are nonnegative,

$$\tau_{\text{scale}}(A) = n(2u + u^2)D(A), \qquad \tau_{\text{dot}}(A) = n\gamma_s(1+u)^2 D(A).$$

These are worst-case Frobenius budgets for, respectively, rounded row scaling and rounded dot products in the Gram computation.

Theorem 7 (Rounded scaling, exact Gram summation)**.** *Assume $D(A) > 0$, $s > 0$, and $\varepsilon > 0$. If the sketch entries are rounded but $G(\widehat{\widetilde{A}}) = \widehat{\widetilde{A}}^T\widehat{\widetilde{A}}$ is formed as an exact real Gram matrix, then*

$$\mathbb{P}\left[\left\|\widehat{G}_{\text{scale}} - A^T A\right\|_F \le \varepsilon\, D(A) + \tau_{\text{scale}}(A)\right] \ge 1 - \frac{1}{s\varepsilon^2}.$$

Lean formalization.

`rowSampleGramFpPerturbBudget`

`rowSampleGramFpPerturbBudget_eq_nat_mul`

`rowSampleGram_perturb_budget_le_explicit`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_explicit_budget`

Theorem 8 (Fully floating-point Gram approximation)**.** *Assume $D(A) > 0$, $s > 0$, $\varepsilon > 0$, and `gammaValid fp s`. If each Gram entry is computed by the repository's floating-point dot-product algorithm after Algorithm 2 has already returned its sampled-and-scaled rows, then*

$$\mathbb{P}\left[\left\|\widehat{G}_{\text{dot}} - A^T A\right\|_F \le \varepsilon\, D(A) + \tau_{\text{full}}(A)\right] \ge 1 - \frac{1}{s\varepsilon^2}.$$

Lean formalization.

`dotProduct_error_bound`

`rowSampleGramDotProductBudget`

`rowSampleGramDotProductBudget_eq_nat_mul`

`rowSampleGramFullFpPerturbBudget`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGramDot_frob_error_le_epsilon_add_explicit_budget`

Corollary 9 ($\tau_{\text{dot}} = 0$ scaling-only specialization)**.** *Assume $D(A) > 0$, $s > 0$, and $\varepsilon > 0$. If Algorithm 2 only computes the sampled-and-scaled rows in floating point, and the Gram matrix is introduced afterwards as the exact mathematical matrix $\widehat{G}_{\text{scale}} = \widehat{\widetilde{A}}^T\widehat{\widetilde{A}}$, then the appropriate analysis model has zero dot-product budget:*

$$\tau_{\text{dot}}(A) = 0.$$

Consequently,

$$\mathbb{P}\left[\left\|\widehat{G}_{\mathrm{scale}} - A^T A\right\|_F \le \varepsilon\, D(A) + \tau_{\mathrm{scale}}(A)\right] \ge 1 - \frac{1}{s\varepsilon^2},$$

where

$$\tau_{\mathrm{scale}}(A) = n(2u + u^2)D(A).$$

This is formalized as a separate theorem about the theoretical Gram matrix formed from rounded rows, not as the computed-dot-product matrix with an ignored dot-product error. It is the faithful Algorithm 2 floating-point stability statement when the Gram matrix appears only in the analysis.

Lean formalization.

`rowSampleGramFpPerturbBudget_eq_nat_mul`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_scaling_budget`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_tau_dot_zero`

5 Equation (6): Leverage-Score Sampling

Let $U \in \mathbb{R}^{m\times n}$ have orthonormal columns:

$$U^T U = I_n.$$

The i-th leverage score is

$$\ell_i = \|U_{i*}\|_2^2\,.$$

Since

$$\sum_{i=1}^{m} \ell_i = \|U\|_F^2 = \mathrm{trace}(U^T U) = n,$$

equation (6) is

$$p_i = \frac{\ell_i}{\sum_r \ell_r} = \frac{\ell_i}{n}.$$

Theorem 10 (Formal equation (6))**.** *If $U^T U = I_n$, then the squared-row-norm distribution applied to U satisfies*

$$p_i = \frac{\|U_{i*}\|_2^2}{n} \qquad \text{and} \qquad \sum_i p_i = 1$$

for $n > 0$.

Lean formalization.

`HasOrthonormalColumns`

`leverageScore`

`leverageScoreProb`

`rowSqNormProbDen_eq_nat_of_orthonormal_columns`

`leverageScoreProb_eq_rowNormSq_div_nat`

`leverageScoreProb_sum_eq_one`

6 Equation (7): Subspace Embedding

The repository states the operator-2 estimate in vector-action form. For a square matrix M, define

$$\mathrm{Op2Le}(M, c) \iff \forall x \in \mathbb{R}^n,\ \|Mx\|_2 \le c\,\|x\|_2\,.$$

The formal proof uses the standard implication

$$\|M\|_F \le c \implies \mathrm{Op2Le}(M, c).$$

Lean formalization.

`vecNorm2`

`opNorm2Le`

`vecNorm2_matMulVec_le_frobNorm_mul`

`opNorm2Le_of_frobNorm_le`

Theorem 11 (Equation (7), exact arithmetic)**.** *Let $U \in \mathbb{R}^{m\times n}$ satisfy $U^TU = I_n$. Assume $n > 0$, $s > 0$, and $\varepsilon > 0$. Let $\widetilde{U}$ be the Algorithm 2 sketch produced by sampling rows of U with the leverage probabilities $p_i = \|U_{i*}\|_2^2/n$. Then*

$$\mathbb{P}\Big[\forall x \in \mathbb{R}^n, \Big\|(\widetilde{U}^T\widetilde{U} - I_n)x\Big\|_2 \le \varepsilon\,\|x\|_2\Big] \ge 1 - \frac{1}{s(\varepsilon/n)^2}.$$

Lean formalization.

`rowGram_eq_id_of_orthonormal_columns`.

`leverageTraceProbability_eventProb_rowSampleGram_opNorm2_error_le_epsilon`

Corollary 12 (Equation (7), concrete computed-denominator floating-point arithmetic)**.** *Assume the exact Bennett sample-budget hypotheses for equation (7), and assume `gammaValid fp s`. Let $\widehat{G}_{\mathrm{dot}}(U)$ be the floating-point sampled Gram matrix formed from the exact leverage law $p_i = \|U_{i*}\|_2^2/n$, the concrete denominator routine*

$$\widehat{d_i} = \mathrm{flsqrt}(\mathrm{flmul}(s, p_i)),$$

rounded row divisions, and rounded length-s Gram dot products. Then the two-sided finite-Loewner event

$$\widehat{G}_{\mathrm{dot}}(U) - I_n \preceq \big(\varepsilon + \tau_{\mathrm{comp}}(U; s)\big)I_n, \qquad I_n - \widehat{G}_{\mathrm{dot}}(U) \preceq \big(\varepsilon + \tau_{\mathrm{comp}}(U; s)\big)I_n$$

has probability at least $1 - \delta$, with

$$\tau_{\mathrm{comp}}(U; s) = n^2\left((2\eta_U + \eta_U^2) + \gamma_s(1 + \eta_U)^2\right).$$

Here η_U is the effective row-scaling relative error induced by the same computed denominator routine. The matrix U is an exact supplied orthonormal-column basis; this corollary does not compute U from a data matrix.

Lean formalization.

`leverage_fl_rowSampleGramDot_perturb_bound`.

`leverageTraceProbability_eventProb_fl_rowSampleGramDot_opNorm2_error_le_epsilon_add_budget`

`leverageFlMulThenSqrtRowScaleDen`

`leverageTraceProbability_eventProb_fl_rowSampleGramDotWithFlMulThenSqrtDen_two_sided_`
`finiteLoewnerLe_ge_one_sub_delta_of_sample_budget`

Corollary 13 (Equation (7), actual input $A = UC$ and computed Gram)**.** *Let $U \in \mathbb{R}^{m\times r}$ satisfy $U^T U = I_r$, let $C \in \mathbb{R}^{r\times q}$, and set $A = UC$. The exact leverage law is $p_i = \|U_{i*}\|_2^2/r$, and the implementation samples rows of the actual matrix A. The exact analysis witnesses U, C are not computed by this theorem.*

Under the same Bennett sample-budget hypotheses with r in place of n, the exact sampled actual-input Gram satisfies

$$\Pr\Big[\widetilde{A}_U^T\widetilde{A}_U - A^TA \preceq \varepsilon A^TA \text{ and } A^TA - \widetilde{A}_U^T\widetilde{A}_U \preceq \varepsilon A^TA\Big] \geq 1-\delta.$$

With the concrete denominator $\widehat{d}_i = \mathrm{fl}_{\mathrm{sqrt}}(\mathrm{fl}_{\mathrm{mul}}(s, p_i))$, rounded row divisions on entries of A, and rounded length-s Gram dot products, the computed Gram satisfies

$$\Pr\Big[\widehat{G}_{\mathrm{dot}}(A;U) - A^TA \preceq \varepsilon A^TA + T_A^{\mathrm{fp}} I_q \text{ and } A^TA - \widehat{G}_{\mathrm{dot}}(A;U) \preceq \varepsilon A^TA + T_A^{\mathrm{fp}} I_q\Big] \geq 1-\delta.$$

Here

$$T_A^{\mathrm{fp}} = \big(\gamma_s(1+\eta_U^A)^2 + 2\eta_U^A + (\eta_U^A)^2\big)\sum_{j=1}^{q}\left(\sum_{i:\, p_i>0}\frac{|A_{ij}|}{\sqrt{p_i}}\right)^2,$$

where

$$\eta_U^A = u + (1+u)\sum_{i:\, p_i>0}\frac{\sqrt{sp_i}\,(\sqrt{1+u}\,u+u)}{|\,\mathrm{fl}_{\mathrm{sqrt}}(\mathrm{fl}_{\mathrm{mul}}(s,p_i))|}.$$

For fixed U, C, m, r, q, s, this radius is

$$\Theta\left((\gamma_s + u(1+|\{i : p_i>0\}|))\sum_{j=1}^{q}\left(\sum_{i:\, p_i>0}\frac{|A_{ij}|}{\sqrt{p_i}}\right)^2\right)$$

and tends to zero as $u, \gamma_s \to 0$ when positive-probability computed denominators remain nonzero.

Lean formalization.

`leverageFactoredInputSampleGram`

`leverageTraceProbability_eventProb_factoredInputSampleGram_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget`

`leverageFactoredInputDenGramBudget`

`leverage_fl_rowSampleGramDotWithComputedDen_factoredInput_perturb_bound`

`leverageTraceProbability_eventProb_factoredInput_fl_rowSampleGramDotWithFlMulThenSqrtDen_two_sided_finiteLoewnerLe_ge_one_sub_delta_of_sample_budget`

Corollary 14 (Equation (7), stored basis table and computed Gram)**.** *Let $U \in \mathbb{R}^{m\times n}$ satisfy $U^TU = I_n$, and let the exact leverage law be $p_i = \|U_{i*}\|_2^2/n$. The probabilities and samples are exact mathematical objects. The three closed stored-basis modes are*

$$\widehat{U}_{ij}^{(\mathrm{mul1})} = \mathrm{fl}_{\mathrm{mul}}(U_{ij}, 1), \qquad \widehat{U}_{ij}^{(\mathrm{add0})} = \mathrm{fl}_{\mathrm{add}}(U_{ij}, 0), \qquad \widehat{U}_{ij}^{(\mathrm{sub0})} = \mathrm{fl}_{\mathrm{sub}}(U_{ij}, 0),$$

and all satisfy

$$|\widehat{U}_{ij}^{(\mu)} - U_{ij}| \leq u|U_{ij}|.$$

With

$$\rho_U = \sum_{i:\, p_i>0} \frac{\sqrt{sp_i}\,(\sqrt{1+u}\,u+u)}{|\,\mathrm{fl}_{\mathrm{sqrt}}(\mathrm{fl}_{\mathrm{mul}}(s,p_i))|}, \qquad \lambda_U = 2u+u^2+(1+u)^2\rho_U,$$

define the expanded deterministic radius

$$T_U^{\mathrm{fp}} = \big(\gamma_s(1+\lambda_U)^2 + 2\lambda_U + \lambda_U^2\big) \sum_{j=1}^{n} \left(\sum_{i:\, p_i>0} \frac{|U_{ij}|}{\sqrt{p_i}} \right)^2 .$$

Under the exact Bennett sample-budget hypotheses and γ_s validity, the Gram matrix computed from $\widehat{U}^{(\mu)}$, the concrete denominator $\widehat{d_i} = \mathrm{fl}_{\mathrm{sqrt}}(\mathrm{fl}_{\mathrm{mul}}(s,p_i))$, rounded row divisions, and rounded length-s dot products satisfies

$$\Pr\Big[\widehat{G}^{(\mu)}_{\mathrm{dot}} - I_n \preceq (\varepsilon + T_U^{\mathrm{fp}})I_n \text{ and } I_n - \widehat{G}^{(\mu)}_{\mathrm{dot}} \preceq (\varepsilon + T_U^{\mathrm{fp}})I_n\Big] \geq 1-\delta$$

for every $\mu \in \{\mathrm{mul1}, \mathrm{add0}, \mathrm{sub0}\}$. This charges the stored basis table, denominator formation, row divisions, and Gram dot products. It does not claim that a QR or SVD routine has computed U from a data matrix.

Lean formalization.

`RowSamplingLeverageComputedBasis.lean`

`leverageComputedBasisDenGramBudget`

Stored-basis final wrappers are provided for the mul-one, add-zero, and sub-zero modes.

7 Hypotheses and Scope

- The exact equation (7) theorem assumes only $U^T U = I_n$, $n > 0$, $s > 0$, and $\varepsilon > 0$.
- The source-aligned finite-Loewner equation (7) theorem assumes the displayed Bennett sample-budget inequalities and $n > 1$.
- The floating-point equation (7) corollaries additionally assume `gammaValid fp s`, inherited from the dot-product theorem.
- The older vector-action/Frobenius-to-operator consequence remains available, but the final equation (7) theorem surface is the formalized Oliveira/Tropp-style finite-Loewner Bennett route.
- The actual-input theorem uses exact analysis witnesses $A = UC$ and computes with rows of A; it charges the concrete denominator, row divisions, and Gram dot products.
- The stored-basis theorem charges the modeled rounded-copy table used in the sketch. It still does not formalize a QR, SVD, or other routine that generates U from a data matrix.
- The proof does not formalize computation or approximation of leverage probabilities. Equation (6) is the exact input sampling distribution once an orthonormal-column matrix U is fixed.
- The proof uses the vector-action operator predicate Op2Le rather than introducing a supremum-valued spectral norm object.

8 File Map

- `RowSampling.lean`: row probabilities, product traces, sampled row scaling, and local floating-point scaling stability.
- `RowSamplingGram.lean`: Gram matrices, unbiasedness, second moments, equation (5), and the floating-point Gram perturbation theorems.
- `RowSamplingLeverage.lean`: equation (6) and equation (7) for leverage-score row sampling.
- `RowSamplingLeverageMGF.lean`: source-aligned Bennett finite-Loewner equation (7), including the concrete computed-denominator floating-point endpoint.
- `RowSamplingLeverageComputedBasis.lean`: actual-input $A = UC$ equation (7) endpoint and stored-basis table equation (7) endpoint for the three rounded-copy modes $\mathrm{fl}_{\mathrm{mul}}(U_{ij}, 1)$, $\mathrm{fl}_{\mathrm{add}}(U_{ij}, 0)$, and $\mathrm{fl}_{\mathrm{sub}}(U_{ij}, 0)$.
- `MatrixAlgebra.lean`: Frobenius norm, vector 2-norm, and the Frobenius-to-operator bound used in equation (7).
- `DotProduct.lean`: floating-point dot-product stability reused by the fully floating-point Gram theorem.
- `FiniteProbability.lean`: finite Markov, union-bound, monotonicity, and expectation lemmas.